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Artificial Intelligence

CO2_AT1

1. Question 1 – Breadth-First Search (BFS)

Consider the following graph:



Task: Starting from node **A**, perform a **Breadth-First Search (BFS)**. Complete the following table.

Iteration	Queue	Visited Node
1		
2		
3		
4		
5		
6		
7		

Questions:

1. Write the BFS traversal order.
2. Show the queue contents after each iteration.

ANSWERS:

1.BFS traversal order:

$$\boxed{A \rightarrow B \rightarrow C \rightarrow D \rightarrow E \rightarrow F \rightarrow G}$$

2. **Queue contents after each iteration:**

*Iteration 1: Start at A. Visit A and add its children B and C to the queue. Queue: [B, C].

*Iteration 2: Remove B and visit it. Add its children D and E. Queue: [C, D, E].

*Iteration 3: Remove C and visit it. Add its children F and G. Queue: [D, E, F, G].

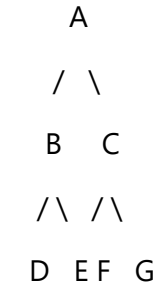
*Iteration 4: Remove D and visit it. D has no children. Queue: [E, F, G].

*Iteration 5: Remove E and visit it. E has no children. Queue: [F, G].

*Iteration 6: Remove F and visit it. F has no children. Queue: [G].

* Iteration 7: Remove G and visit it. G has no children. Queue becomes empty, so BFS stops.

2. Depth-First Search (DFS) Using the same graph,



Perform **Depth-First Search (DFS)** starting from **A**.

Complete the table.

ANSWER:

1. **DFS traversal order**

$$\boxed{A \rightarrow B \rightarrow D \rightarrow E \rightarrow C \rightarrow F \rightarrow G}$$

2. **Step-by-step explanation**

    1. **Iteration 1: Start at A and visit it. Explore B first.**
    2. **Iteration 2: Visit B. Explore its left child D.**
    3. **Iteration 3: Visit D. It has no children, so backtrack to B.**
    4. **Iteration 4: Visit E, the remaining child of B. Then backtrack to A.**
    5. **Iteration 5: Visit C and explore its left child F.**

6. **Iteration 6: Visit F. It has no children, so backtrack to C.**
7. **Iteration 7: Visit G. All nodes have now been visited.**

3. **DFS iteration table**

Iteration	**Stack (Top → Bottom)**	**Visited Node**
1	B, C	A
2	D, E, C	B
3	E, C	D
4	C	E
5	F, G	C
6	G	F
7	Empty	G

3. **Initial State :**

1 3 6

5 0 2

4 7 8

Goal State :

1 2 3

4 5 6

7 8 0

Task

Trace **Breadth-First Search (BFS)** until the goal state is reached.

Fill in the table.

Iteration	Current State	Queue Before Expansion	Queue After Expansion
1			
2			
3			
4			
5			

Questions

1. Which node is expanded in each iteration?
2. How many states are generated in total?
3. Write the complete solution path from the initial state to the goal state.
4. Why does BFS always find the shortest solution in an unweighted 8-puzzle problem?

ANSWER:

1. Which node is expanded in each iteration?

The first five expanded states are S0, S1, S2, S3 and S4, in that order.

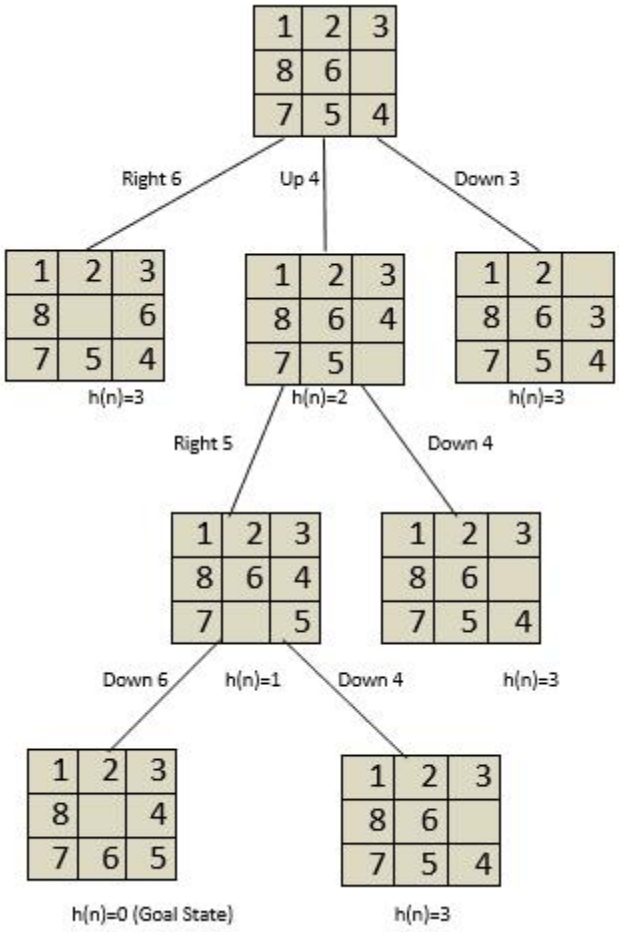
2. How many states are generated in total?

Using the specified move order (Up → Down → Left → Right) and avoiding duplicate states, 500 unique states are discovered by the time the goal is removed from the queue. The goal is reached at iteration 311.

This count depends on the move order and duplicate-handling convention.

3. Complete solution path from initial state to goal state

The shortest solution requires 8 moves. The following path uses the blank tile (0):



4. Why does BFS always find the shortest solution in an unweighted 8-puzzle problem?

BFS explores all states at depth 0, then depth 1, depth 2, and so on. Since every legal move has the same cost of 1, the first time BFS reaches the goal, it has found a solution with the minimum possible number of moves.